SignalDNA

Learning neural networks from first principles

78.78%
Held-out accuracy

0.8613
Held-out ROC-AUC

67,713
Network parameters

THE EXPERIMENT

Built to learn neural networks from first principles and explore future FPGA and embedded engineering applications, SignalDNA uses a NumPy reference network to classify synthetic transients. It examines generalization, numerical equivalence and CPU/GPU performance.

NETWORK ARCHITECTURE

1024 inputs > 64 ReLU > 32 ReLU > 1 sigmoid

12,000 balanced signals | 1,800 held-out | 2,048 Hz | 0.5 s | 60 epochs

01 / TRAINING AND GENERALIZATION

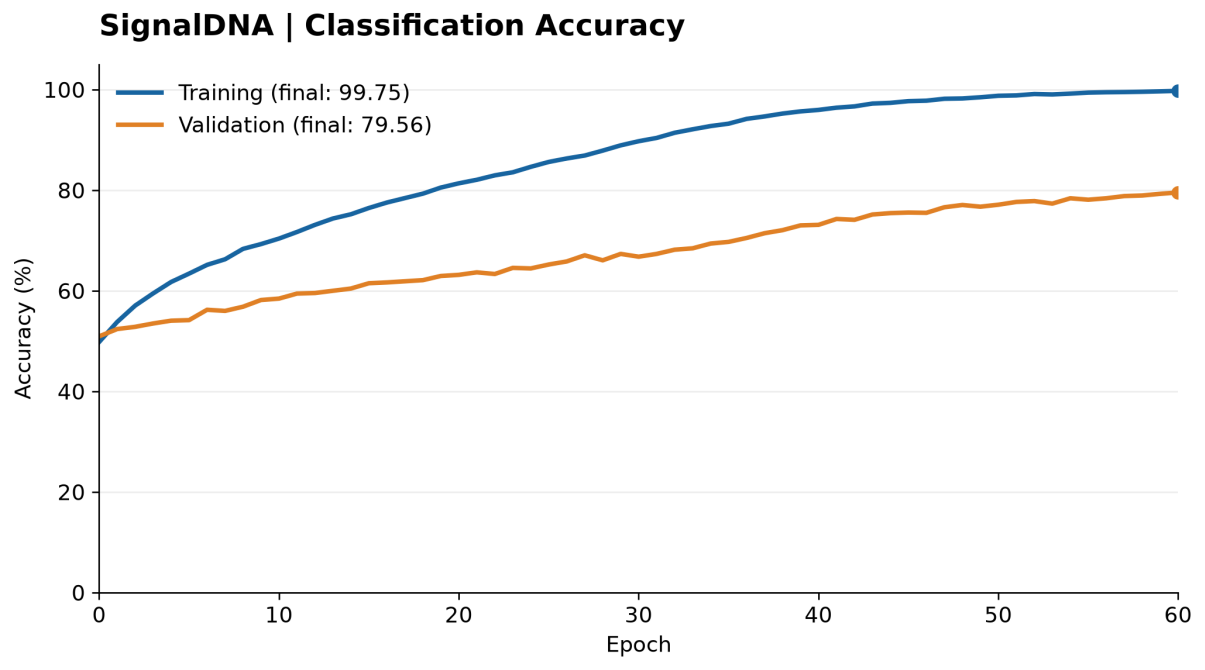


Figure 1. Archived training and validation accuracy, epochs 0-60.

Training 99.75% / Validation 79.56% / Test 78.78%

A substantial generalization gap remains despite high training accuracy.

What the experiments revealed

Reproducibility, unexpected outcomes, and engineering lessons

02 / CPU VERSUS CUDA

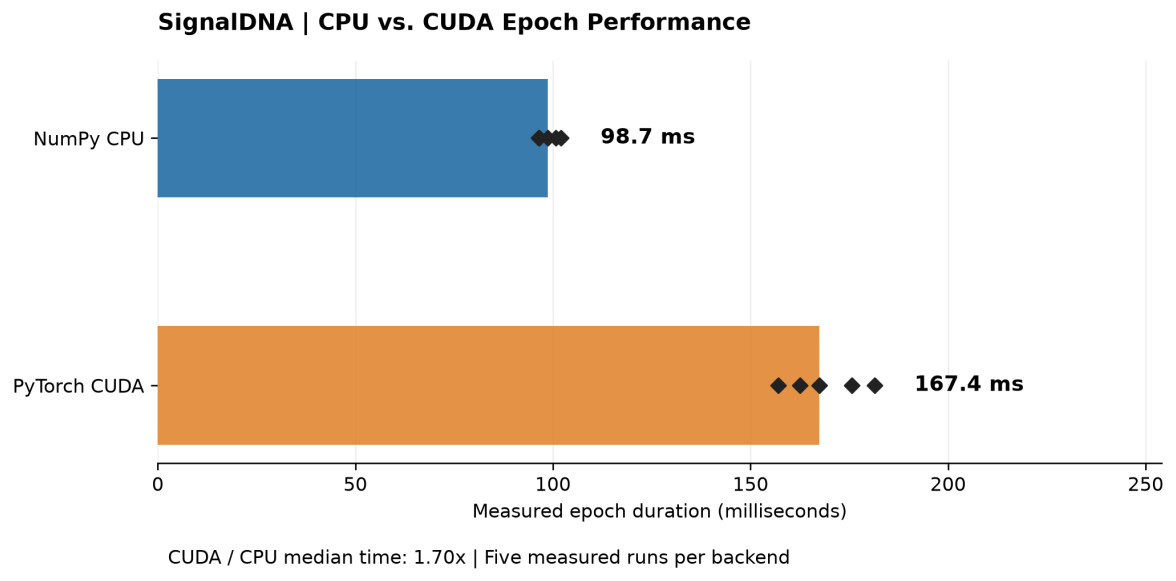


Figure 2. Five warmed-up runs; float64 and batch size 64.

98.72 ms
CPU median epoch

167.37 ms
CUDA median epoch

1.695x
CUDA / CPU time

NUMERICAL EQUIVALENCE

CPU and CUDA learning metrics matched across 61 archived checkpoints. Maximum absolute metric difference: 2.220e-16. Both harnesses shared NumPy metric evaluation.

WHAT WE LEARNED

GPU acceleration is not automatically faster. This specific small-workload benchmark favored CPU execution. The finding does not establish general CPU superiority.

RESEARCH LIMITATIONS

Synthetic data only; one principal architecture and seed; a single locked held-out evaluation. No real-sensor, embedded, or FPGA deployment has been demonstrated.

NEXT RESEARCH DIRECTIONS

Classical signal baselines, regularization, real sensor signals, cross-seed analysis, quantization, FPGA inference, latency, resource use, and power measurements.